

Dae Hyun Lee

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Summary

Data-focused engineer with hands-on experience building end-to-end ML pipelines, production data systems, and large-scale analytics applications. Skilled in Python, SQL, and Spark; proven track record of delivering deployed, data-driven products from raw data ingestion through visualization.

EDUCATION

University of Washington | *M.S., Data Science*

Mar 2027

University of Washington | *B.A., Mathematics*

Mar 2025

Honors & Awards

Databricks - UW Hackathon -- First Place | *TerraCast*

Feb 2025

- Designed and led end-to-end LightGBM optimization pipeline on Databricks using Spark and Python for county-level crop yield prediction; engineered 45+ features from NOAA weather and regional datasets, applied crop-specific modeling and yield detrending to isolate pure weather signal from long-term trend
- Identified and eliminated data leakage sources through rigorous feature validation to ensure unbiased model evaluation; implemented z-score-based anomaly detection to flag performance deviations relative to expected baselines; integrated Databricks Genie (LLM-powered assistant) for natural-language querying of model outputs
- <https://github.com/HumfDev/TerraCast>

Databricks Student Fellowship Program | *Selected as 1 of 39 (top 0.8%) from over 5,000 applicants*

May 2026 - Present

WORK EXPERIENCE

Roy Lab, University of Washington | *Research Assistant*

Oct 2025 - Present

- Developed and benchmarked deep learning architectures (3D CNN, ViT, ResNet, ResNeXt) on multimodal neuroimaging data (4,508+ subjects, 127 features) for three-way Alzheimer's classification (CN/MCI/AD), improving accuracy from 55% to 70% through systematic architecture search and rigorous cross-validated evaluation
- Designed patient-wise data splitting, class-balanced sampling, and 5-fold CV inference pipeline to extract a latent cognitive reserve signal from model prediction heterogeneity; reframed residuals as clinically meaningful features targeting precision medicine applications
- Managed full experiment lifecycle using WandB for tracking and automated Google Sheets logging; leading manuscript preparation on cognitive reserve measurement methodologies in neuroimaging (submission pending)

SNU VLDB Lab, Seoul National University | *Research Intern*

Jun 2024 - Sep 2024

- Designed and validated a z-score-based dimensionality reduction pipeline for embedding vectors, achieving 96% size reduction (1,536 to 64 dimensions) while preserving 95%+ similarity accuracy
- Integrated optimized algorithm into HNSW indexing system; benchmarked retrieval quality and latency across 100K+ embedding vectors and authored internal technical reports documenting results

PROJECTS

CarePilot | [GitHub](#) | *Evidence-aware healthcare facility referral copilot*

Jun 2026 - Jun 2026

- Built CarePilot, an AI chatbot that recommends the most suitable hospital based on the user's situation, using an end-to-end Databricks stack (Lakebase, Genie, Model Serving)
- Designed a dynamic weighted scoring system that re-ranks hospitals in real time (e.g. distance weight increases with case severity) and provides driving routes via OSRM

WatchDawg | [WatchDawg](#) | *Seattle Crime Analytics Dashboard*

Sep 2025

- Built production analytics dashboard on 500K+ Seattle PD incidents (17+ years) with real-time filtering and geospatial visualization; engineered 11-stage data cleaning pipeline reducing dataset noise by 35%

Lillasy | [lillasy.com](#) | *Skincare Recommendation*

Jan 2026

- Built ingredient-based recommendation engine over 4,500+ skincare products; scored products by matching ingredient profiles to user skin type, penalizing flagged ingredients using Paula's Choice dictionary
- Deployed as live web service at lillasy.com with CI/CD pipeline (Ruff, pytest, Codecov); implemented KNN-based retrieval with prebuilt JSON caches for low-latency inference

Optimizing Clean Water Supply | [GitHub](#) | *EY Competition - 2026*

Nov 2025 - Mar 2026

- Designed and implemented a spatially aware ML pipeline for the EY 2026 Clean Water Supply Challenge, engineering remote-sensing and climate features (Landsat, TerraClimate, ERA5, HydroRIVERS) and using two-stage modeling for DRP to improve robustness under distribution shift

TECHNICAL SKILLS

- Languages:** Python, Java, R, SQL
- ML/DL Frameworks:** Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, Random Forest
- Machine Learning Skills:** End-to-End ML Pipelines, Optimization, Simulation, Supervised & Unsupervised Learning
- Others:** Spark, Pandas, NumPy, Matplotlib, Seaborn, Tableau, Dash, Plotly, Google Cloud, Excel, Git